

FMUSER FUQ-230 Modular Headend Processor 128 IP Inputs 3 Module Slots DVB-C-DVB-T-ATSC-ISDB-T ASI-S2X-USB Multi-Standard Demodulation Remultiplexing for Terrestrial Cable Satellite Networks Product overview

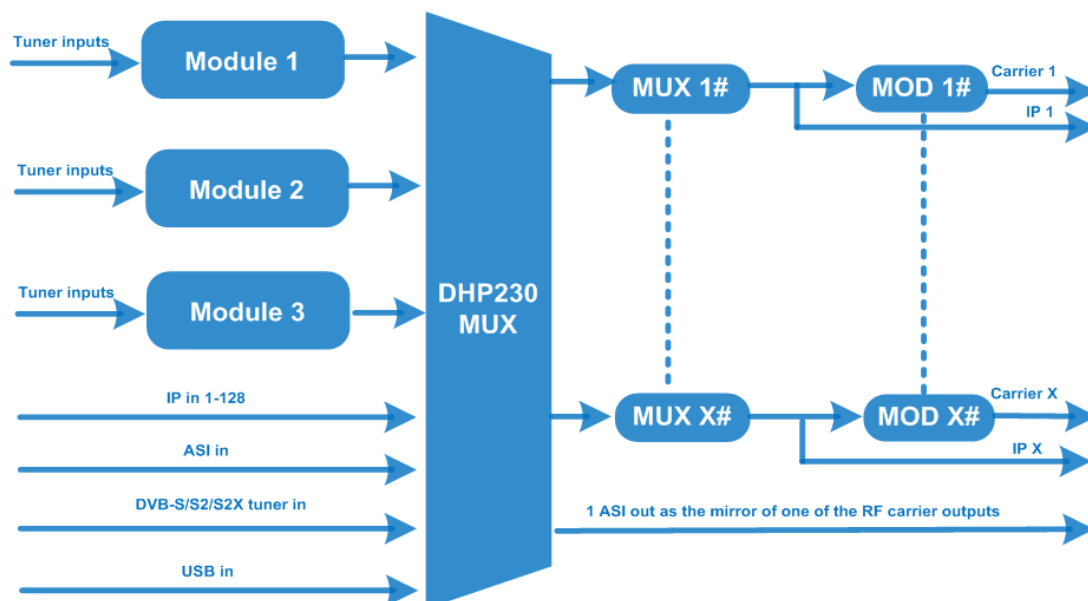


The FMUSER FUQ-230 is a 1U modular headend processor with 128 IP inputs, 1 ASI, 1 DVB-S/S2/S2X, and 1 USB interface. It processes signals via demodulation, descrambling, and remultiplexing using 3 swappable modules. Web-based NMS control enables remote management for cable headends, terrestrial transmitters, and IPTV distribution.

Key feature

MPTS over UDP/RTP/RTSP with 256 PIDs per channel

Principle Chart



Chassis Specifications

Input	3 Modules inputs(see the module specification) 1 ASI in for re-multiplexing 128 IP(SPTS/MPTS) inputs over UDP and RTP, RJ45, 1 000M/100M D ATA 1 , Unicast/Multicast 1 DVB-S/S2/S2 X tuner in for re-multiplexing 1 USB Player input for re-mux ("xxx.ts" video)	3 Modules inputs(see the module specification) 1 ASI in for re-multiplexing 128 IP(SPTS/MPTS) inputs over UDP and RTP, RJ45, 1 000M/100M D ATA 1 , Unicast/Multicast 1 DVB-S/S2/S2 X tuner in for re-multiplexing 1 USB Player input for re-mux ("xxx.ts" video)	3 Modules inputs(see the module specification) 1 ASI in for re-multiplexing 128 IP(SPTS/MPTS) inputs over UDP and RTP, RJ45, 1 000M/100M D ATA 1 , Unicast/Multicast 1 DVB-S/S2/S2 X tuner in for re-multiplexing 1 USB Player input for re-mux ("xxx.ts" video)	3 Modules inputs(see the module specification) 1 ASI in for re-multiplexing 128 IP(SPTS/MPTS) inputs over UDP and RTP, RJ45, 1 000M/100M D ATA 1 , Unicast/Multicast 1 DVB-S/S2/S2 X tuner in for re-multiplexing 1 USB Player input for re-mux ("xxx.ts" video)	3 Modules inputs(see the module specification) 1 ASI in for re-multiplexing 128 IP(SPTS/MPTS) inputs over UDP and RTP, RJ45, 1 000M/100M D ATA 1 , Unicast/Multicast 1 DVB-S/S2/S2 X tuner in for re-multiplexing 1 USB Player input for re-mux ("xxx.ts" video)
Multiplexing	Maximum PID Remapping	256 input per channel	256 input per channel	256 input per channel	256 input per channel
Multiplexing	Function	A/V PID Filtering	A/V PID Filtering	A/V PID Filtering	A/V PID Filtering
Multiplexing	Function	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)
Multiplexing	Function	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting
Multiplexing	Function	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically
Modulation	DVB-C	QAM Channel	QAM Channel	16 non-adjacent carriers output (maximum bandwidth 192MHz)	16 non-adjacent carriers output (maximum bandwidth 192MHz)

Modulation	DVB-C	Standard	Standard	EN300 429/ITU-T J.83A/B	EN300 429/ITU-T J.83A/B
Modulation	DVB-C	MER	MER	≥40db	≥40db
Modulation	DVB-C	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulation	DVB-C	RF output level	RF output level	-20~+3dbm, 0.1db step	-20~+3dbm, 0.1db step
Modulation	DVB-C	Symbol Rate	Symbol Rate	5.0Msps~7.0 Msps, 1ksps stepping	5.0Msps~7.0 Msps, 1ksps stepping
Modulation	DVB-C			J.83A	J.83B
Modulation	DVB-C	Constellation	Constellation	16/32/64/128/ 256QAM	64/256 QAM
Modulation	DVB-C	Bandwidth	Bandwidth	8M	6M
Modulation	DVB-T	Standard	Standard	EN300744	EN300744
Modulation	DVB-T	FFT mode	FFT mode	2K,4K, 8K	2K,4K, 8K
Modulation	DVB-T	Bandwidth	Bandwidth	6M, 7M, 8M	6M, 7M, 8M
Modulation	DVB-T	Constellation	Constellation	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
Modulation	DVB-T	Guard Interval	Guard Interval	1/4, 1/8, 1/16, 1/32	1/4, 1/8, 1/16, 1/32
Modulation	DVB-T	FEC	FEC	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Modulation	DVB-T	MER	MER	≥42 dB	≥42 dB
Modulation	DVB-T	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulation	DVB-T	RF out	RF out	8 non-adjacent carriers output (maximum bandwidth 192MHz)	8 non-adjacent carriers output (maximum bandwidth 192MHz)
Modulation	DVB-T	RF output level	RF output level	-20~ +3dBm, 0.1db step	-20~ +3dBm, 0.1db step
Modulation	ATSC	Standard	Standard	ATSC A/53	ATSC A/53

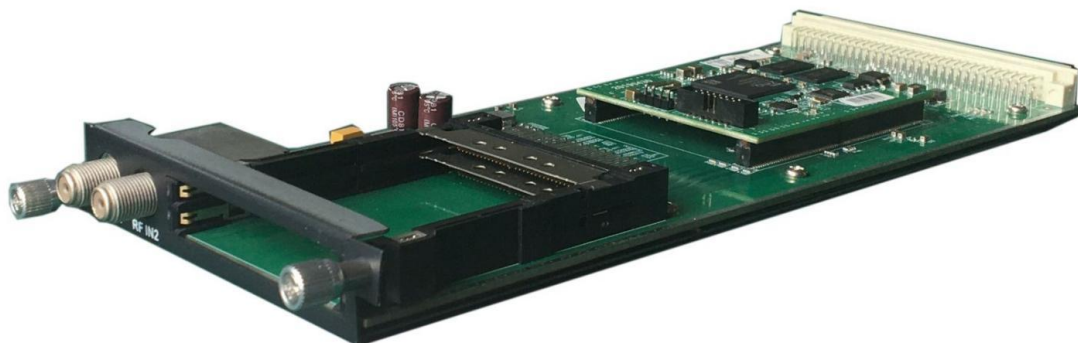
Modulation	ATSC	Bandwidth	Bandwidth	6M	6M
Modulation	ATSC	Constellation	Constellation	8VSB	8VSB
Modulation	ATSC	FEC	FEC	RS(208 188)+Trellis	RS(208 188)+Trellis
Modulation	ATSC	MER	MER	≥40dB	≥40dB
Modulation	ATSC	ACL	ACL	-55 dBc	-55 dBc
Modulation	ATSC	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulation	ATSC	RF out	RF out	8 non-adjacent carriers output (maximum bandwidth 192MHz)	8 non-adjacent carriers output (maximum bandwidth 192MHz)
Modulation	ATSC	RF output level	RF output level	-20~+3dbm (f or all carriers), 0.5db stepping	-20~+3dbm (f or all carriers), 0.5db stepping
Modulation	ISDB-T	Standard	Standard	ARIB STD-B31	ARIB STD-B31
Modulation	ISDB-T	Bandwidth	Bandwidth	6M	6M
Modulation	ISDB-T	Constellation	Constellation	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
Modulation	ISDB-T	Guard Interval	Guard Interval	1/32, 1/16, 1/8, 1/4	1/32, 1/16, 1/8, 1/4
Modulation	ISDB-T	Transmission Mode	Transmission Mode	2K, 4K, 8K	2K, 4K, 8K
Modulation	ISDB-T	Code rate	Code rate	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Modulation	ISDB-T	MER	MER	≥40dB	≥40dB
Modulation	ISDB-T	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step

Modulation	ISDB-T	RF out	RF out	6 non-adjacent carriers output (maximum bandwidth 192MHz)	6 non-adjacent carriers output (maximum bandwidth 192MHz)
Modulation	ISDB-T	RF output level	RF output level	-20dBm~+3dBm, 0.1dB stepping	-20dBm~+3dBm, 0.1dB stepping
Stream output	1 ASI output as mirror of one of RF output carriers 16 MPTS output over UDP and RTP/RTSP as mirror of 16 DVB-C carriers, 8 MPTS output over UDP and RTP/RTSP as mirror of 8 DVB-T/ATSC carriers(Optional) 6 MPTS output over UDP and RTP/RTSP as mirror of 6 ISDB-T carriers(Optional) 1 *100M/1000M Base-T Ethernet interface, DATA 2	1 ASI output as mirror of one of RF output carriers 16 MPTS output over UDP and RTP/RTSP as mirror of 16 DVB-C carriers, 8 MPTS output over UDP and RTP/RTSP as mirror of 8 DVB-T/ATSC carriers(Optional) 6 MPTS output over UDP and RTP/RTSP as mirror of 6 ISDB-T carriers(Optional) 1 *100M/1000M Base-T Ethernet interface, DATA 2	1 ASI output as mirror of one of RF output carriers 16 MPTS output over UDP and RTP/RTSP as mirror of 16 DVB-C carriers, 8 MPTS output over UDP and RTP/RTSP as mirror of 8 DVB-T/ATSC carriers(Optional) 6 MPTS output over UDP and RTP/RTSP as mirror of 6 ISDB-T carriers(Optional) 1 *100M/1000M Base-T Ethernet interface, DATA 2	1 ASI output as mirror of one of RF output carriers 16 MPTS output over UDP and RTP/RTSP as mirror of 16 DVB-C carriers, 8 MPTS output over UDP and RTP/RTSP as mirror of 8 DVB-T/ATSC carriers(Optional) 6 MPTS output over UDP and RTP/RTSP as mirror of 6 ISDB-T carriers(Optional) 1 *100M/1000M Base-T Ethernet interface, DATA 2	1 ASI output as mirror of one of RF output carriers 16 MPTS output over UDP and RTP/RTSP as mirror of 16 DVB-C carriers, 8 MPTS output over UDP and RTP/RTSP as mirror of 8 DVB-T/ATSC carriers(Optional) 6 MPTS output over UDP and RTP/RTSP as mirror of 6 ISDB-T carriers(Optional) 1 *100M/1000M Base-T Ethernet interface, DATA 2
System function	Network management (WEB)	Network management (WEB)	Network management (WEB)	Network management (WEB)	Network management (WEB)
System function	Chinese and English language	Chinese and English language	Chinese and English language	Chinese and English language	Chinese and English language

System function	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade
Miscellaneous	Dimension (W×L×H)	Dimension (W×L×H)	482mm×380mm×44mm	482mm×380mm×44mm	482mm×380mm×44mm
Miscellaneous	Environment	Environment	0~45℃(work) -20~80℃(storage)	0~45℃(work) -20~80℃(storage)	0~45℃(work) -20~80℃(storage)
Miscellaneous	Power requirements	Power requirements	AC 110V±10%, 50/60Hz, AC 220 ± 10%, 50/60Hz	AC 110V±10%, 50/60Hz, AC 220 ± 10%, 50/60Hz	AC 110V±10%, 50/60Hz, AC 220 ± 10%, 50/60Hz

Module Specifications

DX902A 2 Tuner Descrambling Module



Stream in:

2 Tuner input, F Type,

Stream out: 1 MPTS output to DHP230 chassis for re-multiplexing

DVB-CI:2 independent common interface slots

Standard: DVB-S/S2/S2X;

DVB-S

Input Frequency: 950-2150MHz

Symbol Rate: QPSK 1~45Msps

Signal Strength: -65~ -25dBm

FEC Demodulation: 1/2, 2/3, 3/4, 5/6, 7/8

DVB-S2

Input Frequency: 950-2150MHz

Symbol rate: QPSK/8PSK 1~45Msps; 16APSK 1~45 Msps; 32APSK 1~32 Msps

FEC Demodulation: 1/2, 2/3, 3/4, 5/6, 7/8, 4/5, 5/6, 8/9, 9/10

DVB-S2X

Input Frequency: 950-2150MHz

Symbol rate: QPSK/8PSK/16APSK 0.5~45 Msps;

8APSK/32APSK: 0.5~40Msps

FEC Demodulation:

QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20

8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10, 23/36, 25/36, 13/18

8APSK: 5/9-L, 26/45-L

16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36, 2/3-L, 25/36, 13/18, 7/9, 77/90

32APSK: 3/4, 4/5, 5/6, 8/9, 2/3-L, 32/45, 11/15, 7/9

Support Diseq function

Multiplexing:

Maximum PID Remapping: 256 output pids

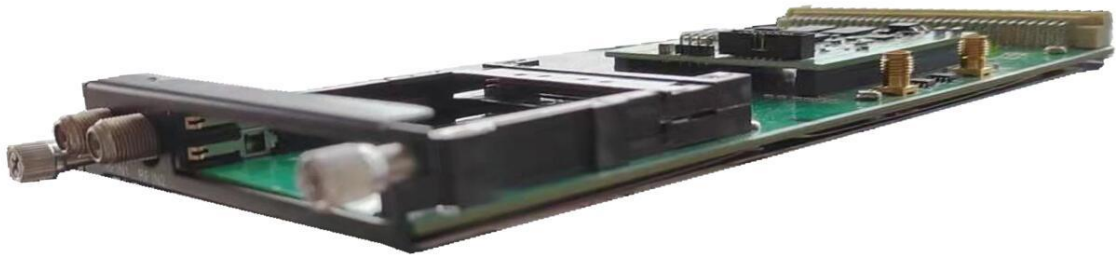
Function: PID remapping (automatically or manually), Accurate PCR adjusting, generate PSI/ SI table automatically

Descrambling:

CAM/CI Quantity: 2

BISS Mode: Mode 1, Mode E; 32 BISS Keys

DX942A 4 frequencies Descrambling Module



Stream in:

4 frequencies input (each RF in interface for 2 frequencies locking), F Type,

Stream out: 1 MPTS output to DHP230 chassis for re-multiplexing

DVB-CI: 2 independent common interface slots

Standard: DVB-C (J.83 A/C)/J.83B/ DVB-T/DVB-T2/ISDB-T switchable

Standard: DVB-C (J.83 A/C); J.83B

Input Frequency: 60MHz~890MHz

Symbol rate: 1000~9000Ksps

Constellation: 16/32/64/128/256 QAM; 64/256 QAM for J.83B

Standard: DVB-T/T2

Frequency In: 60MHz~890MHz

Bandwidth: 5/6/7/8M bandwidth

PLP supported for DVB-T2

Standard: ISDB-T

Input Frequency: 60-890MHz

Multiplexing:

Maximum PID Remapping: 256 output pids

Function: PID remapping (automatically or manually), generate PSI/ SI table automatically

Descrambling:

CAM/CI Quantity: 2

BISS Mode: Mode 1, Mode E; 32 BISS Keys

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More

than 20,000 television stations, radio stations, and hotel clients from different.

Contact:Tomleequan Email■sales@fmuser.com whatsapp: +8613922702227

FMUSER 广州艾曼